Monitoring populations

To compare populations biologists need to look at **densities**. Density refers to the number of individuals of a species per unit area. For instance, species such as kangaroos and weeds can be counted on a per hectare basis. However, it's not possible to count every individual in a population. Individual animals can move or be hidden from view and some weeds grow in clumps or from underground shoots. To combat this, animals can be counted per nesting site or burrow. Clumping plants can be counted as the total weight of the population per unit area. Insects and **microorganisms** might be best expressed as number of individuals per unit of volume. For example water bugs could be counted per litre of pond water.

☞ **1** Read the passage above and underline the different methods that can be used to count populations.

2 Fill in the table below, circling the method you would use to count each population and giving a reason for your answer.

Species	Method of counting	Why
rabbits in a paddock	• individuals per hectare • burrows per hectare	
mosquito larvae in a tank	• individuals per tank • individuals per litre of water	
fish in a large dam	• individuals per square metre of water • kilograms per hectare of water	
thrips in a rose bush	• individuals per flower • grams per rose bush	
capeweed in a garden	• individuals per square metre • kilograms per garden bed	

3 How would you count the number of **broad-leafed** weeds on the school oval or garden beds? _____

_____

4 What is the best measurement of area to use? How do you reach the final number?

_____

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